

Modeling reactive transport of biogenic uraninite re-oxidation by Fe(III)-(hydr)oxides

Benchmark Problem

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1. Benchmark problem set overview

Subsurface bacteria such as sulfate reducing bacteria (SRB) can reduce soluble U(VI) to insoluble U(IV) as uraninite (UO_2), which can lead to U immobilization. On the other hand, it has been shown that once the electron donor is entirely consumed, SRB-mediated uraninite can be reoxidized by Fe(III)-(hydr)oxides, which may potentially impede cleanup efforts where U(VI)-contaminated groundwater is remediated using organic electron donors to precipitate U(IV) as UO_2 . A reaction network was developed by Spycher et al. (2011) based on the experiments of Sani et al. (2004) to simulate U(VI) bioreduction with concomitant UO_2 reoxidation in the presence of hematite or ferrihydrite. The reduction phase considers SRB reduction using lactate with the reductive dissolution of Fe(III) solids, which is set to be microbially mediated as well as abiotically driven by sulfide. It is shown that the oxidation of sulfide by Fe(III) directly competes with UO_2 reoxidation as thermodynamically, Fe(III) oxidizes sulfide preferentially to UO_2 . The re-oxidation of UO_2 is thus shown to depend on the relative rates of UO_2 and sulfide oxidation by Fe(III), as well as to the activity of Fe(II) in solution (Spycher et al., 2011).

The proposed benchmark problem consists of a batch and 2-D multi-component reactive transport model with the reaction network developed by Spycher et al (2011) to investigate the U(VI) bioreduction and its concomitant re-oxidation in the presence of Fe(III)-(hydr)oxides. The problem set consists of 3 parts:

Part I. Numerical simulation of batch experiments using the multicomponent biogeochemical reaction network developed by Spycher et al (2011) and comparison using Toughreact, Phreeqc and PHT3D.

Part II. Numerical simulation of the biogeochemical reaction network of Part I using a 2-D reactive transport model with constant hydrochemical composition and with constant inflow of same U(VI) containing water at the boundary, and temporary slow injection of lactate in the middle of the domain.

Part III. Part II including surface complexation of U(VI).

2. Simulation description overview:

For Part I:

Part I consists of batch simulation of biogeochemical reactions.

Reaction Network Description:

Processes:

- Equilibrium reactions: Aqueous speciation, precipitation/dissolution of solid phases including UO_2 , hematite, siderite, and mackinawite.
- Kinetic reactions: Sulfate bioreduction, Fe(III) bioreduction, U(VI) bioreduction, Fe(III) reduction by HS, U(VI) oxidation by Fe(III), and precipitation/dissolution of sulfur.

Numerical techniques:

- Newton-Raphson method for equilibrium reactions (in Phreeqc)
- Runge-Kutta method for kinetic reactions (in Phreeqc)
- TOUGHREACT uses a sequential iteration approach. The system of mixed equilibrium-kinetic chemical reaction equations is solved on a grid block by grid block basis by Newton-Raphson iteration.

For Parts II-III:

a. Flow model

Processes

- - Isothermal, single phase, Darcy flow, 2D, steady-state, confined

Numerical technique

- - Finite-Differences
- - Preconditioned Conjugate-Gradient Method

Boundary conditions

- Constant head boundary (1st - type) at both ends

b. Transport model

Processes

▪ Advection, dispersion, dual-domain 1st order mass-transfer, 1D multi-component transport

Numerical technique

- Third-order Total Variation Diminishing (TVD) scheme based on the ULTIMATE algorithm

Boundary conditions

- Constant concentration boundary at the upstream end of the domain.

c. Reaction Network Description:

Processes:

- Part II: same as Part I
- Part III: same as Part I + Surface Complexation reactions of U(VI)

Numerical techniques:

- Newton-Raphson method for equilibrium reactions
- Runge-Kutta method for kinetic reactions
- Coupling of equilibrium and kinetic reactions via sequential operator splitting
- Coupling of reactions and advective-dispersive transport via sequential operator-splitting

3. Detailed problem specification:

For Part I:

Part I consists of batch simulation of biogeochemical reactions only.

Reaction Network Description:

Processes:

- Equilibrium reactions: Aqueous speciation, precipitation/dissolution of solid phases including UO_2 , hematite, siderite, and mackinawite.
- Kinetic reactions: Sulfate bioreduction, Fe(III) bioreduction, U(VI) bioreduction, Fe(III) reduction by HS, U(VI) oxidation by Fe(III), and precipitation/dissolution of sulfur.

The reaction network consists of aqueous speciations involving major ions, precipitation/dissolution of solid phases including UO_2 , hematite, siderite, and mackinawite.

Thermodynamic data for aqueous speciation reactions is as follows:

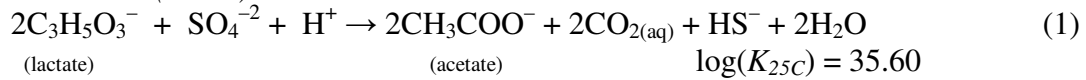
Aqueous speciation reaction	LogK
$\text{H}^+ + \text{Acetate}^- = \text{H}(\text{Acetate})$	4.757
$\text{H}^+ + \text{CO}_3^{2-} = \text{HCO}_3^-$	10.329
$2\text{H}^+ + \text{CO}_3^{2-} = \text{CO}_2 + \text{H}_2\text{O}$	16.681
$\text{Ca}^{2+} + \text{HCO}_3^- = \text{CaCO}_3 + \text{H}^+$	-7.002
$\text{Ca}^{2+} + \text{Cl}^- = \text{CaCl}^+$	-.696

$\text{Ca}^{+2} + 2\text{Cl}^- = \text{CaCl}_2$	-0.644
$\text{Ca}^{+2} + \text{HCO}_3^- = \text{CaHCO}_3^+$	1.047
$\text{Ca}^{+2} + \text{H}_2\text{O} = \text{CaOH}^+ + \text{H}^+$	-12.85
$\text{Ca}^{+2} + \text{Sulfate}^{2-} = \text{CaSulfate}$	2.111
$\text{Fe}^{+2} + 2\text{CO}_3^{2-} = \text{Fe}(\text{CO}_3)_2^{2-}$	7.45
$\text{Fe}^{+2} + 3\text{H}_2\text{O} = \text{Fe}(\text{OH})_3^- + 3\text{H}^+$	-32.962
$\text{Fe}^{+2} + 4\text{H}_2\text{O} = \text{Fe}(\text{OH})_4^{2-} + 4\text{H}^+$	-46
$\text{Fe}^{+3} + 2 \text{Sulfate}^{2-} = \text{Fe}(\text{Sulfate})_2^-$	3.214
$2\text{Fe}^{+3} + 2\text{H}_2\text{O} = \text{Fe}_2(\text{OH})_2^{4+} + 2\text{H}^+$	-7.283
$3 \text{Fe}^{+3} + 4 \text{H}_2\text{O} = \text{Fe}_3(\text{OH})_4^{5+} + 4 \text{H}^+$	-6.3
$\text{Fe}^{+2} + \text{CO}_3^{2-} = \text{FeCO}_3$	5.45
$\text{Fe}^{+2} + \text{Cl}^- = \text{FeCl}^+$	-0.16
$\text{Fe}^{+3} + \text{Cl}^- = \text{FeCl}^{2+}$	1.48
$\text{Fe}^{+2} + 2\text{Cl}^- = \text{FeCl}_2$	-2.45
$\text{Fe}^{+2} + \text{CO}_3^{2-} + \text{H}^+ = \text{FeHCO}_3^+$	11.799
$\text{Fe}^{+2} + \text{H}_2\text{O} = \text{FeO} + 2\text{H}^+$	-20.405
$\text{Fe}^{+3} + \text{H}_2\text{O} = \text{FeO}^+ + 2\text{H}^+$	-5.65
$\text{Fe}^{+3} + 2\text{H}_2\text{O} = \text{FeO}_2^- + 4\text{H}^+$	-21.62
$\text{Fe}^{+2} + \text{H}_2\text{O} = \text{FeOH}^+ + \text{H}^+$	-9.315
$\text{Fe}^{+3} + \text{H}_2\text{O} = \text{FeOH}^{2+} + \text{H}^+$	-4.38
$\text{Fe}^{+2} + \text{HS}^- = \text{FeS} + \text{H}^+$	-2.2
$\text{Fe}^{+2} + \text{Sulfate}^{2-} = \text{FeSulfate}$	2.2
$\text{Fe}^{+3} + \text{Sulfate}^{2-} = \text{FeSulfate}^+$	1.93
$\text{H}^+ + \text{HS}^- = \text{H}_2\text{S}$	7.02
$2\text{H}^+ + \text{Sulfate}^{2-} = \text{H}_2\text{Sulfate}$	0
$\text{Fe}^{+3} + 2\text{H}_2\text{O} = \text{HFeO}_2 + 3\text{H}^+$	-12.018
$\text{Fe}^{+2} + 2\text{H}_2\text{O} = \text{HFeO}_2^- + 3\text{H}^+$	-33.988
$\text{H}^+ + \text{Sulfate}^{2-} = \text{HSulfate}^-$	1.99
$4\text{Mg}^{+2} + 4\text{H}_2\text{O} = \text{Mg}_4(\text{OH})_4^{4+} + 4\text{H}^+$	-39.75
$\text{Mg}^{+2} + \text{CO}_3^{2-} = \text{MgCO}_3$	2.98
$\text{Mg}^{+2} + \text{Cl}^- = \text{MgCl}^+$	-0.1349
$\text{Mg}^{+2} + \text{H}^+ + \text{CO}_3^{2-} = \text{MgHCO}_3^+$	11.339
$\text{Mg}^{+2} + \text{H}_2\text{O} = \text{MgOH}^+ + \text{H}^+$	-11.397
$\text{Mg}^{+2} + \text{Sulfate}^{2-} = \text{MgSulfate}$	2.26
$\text{Na}^+ + \text{CO}_3^{2-} = \text{NaCO}_3^-$	1.27
$\text{Na}^+ + \text{Cl}^- = \text{NaCl}$	-0.777
$\text{Na}^+ + \text{H}^+ + \text{CO}_3^{2-} = \text{NaHCO}_3$	10.079
$\text{Na}^+ + \text{H}_2\text{O} = \text{NaOH} + \text{H}^+$	-14.18
$\text{Na}^+ + \text{Sulfate}^{2-} = \text{NaSulfate}^-$	0.73
$\text{HS}^- = \text{S}^{2-} + \text{H}^+$	-12.918
$11\text{Uranyl}^{+2} + 6\text{CO}_3^{2-} + 12\text{H}_2\text{O} = (\text{Uranyl})_{11}(\text{CO}_3)_6(\text{OH})_{12}^{2-} + 12\text{H}^+$	36.12
$2\text{Uranyl}^{+2} + 2\text{H}_2\text{O} = (\text{Uranyl})_2(\text{OH})_2^{2+} + 2\text{H}^+$	-5.659
$2\text{Uranyl}^{+2} + \text{CO}_3^{2-} + 3\text{H}_2\text{O} = (\text{Uranyl})_2(\text{CO}_3)(\text{OH})_3^- + 3\text{H}^+$	-0.916
$2\text{Uranyl}^{+2} + \text{H}_2\text{O} = (\text{Uranyl})_2\text{OH}^{3+} + \text{H}^+$	-2.729
$3\text{Uranyl}^{+2} + 6\text{CO}_3^{2-} = (\text{Uranyl})_3(\text{CO}_3)_6^{6-}$	53.88
$3\text{Uranyl}^{+2} + 4\text{H}_2\text{O} = (\text{Uranyl})_3(\text{OH})_4^{2+} + 4\text{H}^+$	-11.96
$3\text{Uranyl}^{+2} + 5\text{H}_2\text{O} = (\text{Uranyl})_3(\text{OH})_5^+ + 5\text{H}^+$	-15.62
$3\text{Uranyl}^{+2} + 7\text{H}_2\text{O} = (\text{Uranyl})_3(\text{OH})_7^- + 7\text{H}^+$	-32.2
$3\text{Uranyl}^{+2} + \text{CO}_3^{2-} + 3\text{H}_2\text{O} = (\text{Uranyl})_3\text{O}(\text{OH})_2(\text{HCO}_3)^+ + 3\text{H}^+$	0.583
$4\text{Uranyl}^{+2} + 7\text{H}_2\text{O} = (\text{Uranyl})_4(\text{OH})_7^+ + 7\text{H}^+$	-21.995

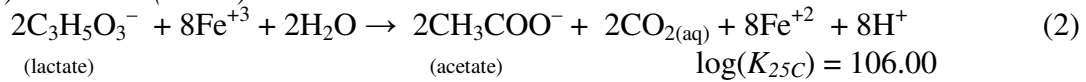
$U^{+4} + 4CO_3^{2-} = U(CO_3)_4^{4-}$	35.05
$U^{+4} + 5CO_3^{2-} = U(CO_3)_5^{6-}$	33.82
$U^{+4} + 2Sulfate^{2-} = U(Sulfate)_2$	10.5
$Uranyl^{+2} + 2Sulfate^{2-} = Uranyl(Sulfate)_2^{2-}$	4.3
$U^{+4} + Cl^- = UCl^{+3}$	1.697
$Uranyl^{+2} + Cl^- = UranylCl^{+}$	0.141
$Uranyl^{+2} + 2Cl^- = UranylCl_2$	-1.146
$Uranyl^{+2} + Sulfate^{2-} = UranylSulfate$	3.18
$Uranyl^{+2} + H_2O = UranylOH^{+} + H^{+}$	-5.217
$2H_2O + Uranyl^{+2} = Uranyl(OH)_2 + 2H^{+}$	-12.152
$Uranyl^{+2} + 3H_2O = Uranyl(OH)_3^{+} + 3H^{+}$	-20.246
$Uranyl^{+2} + 4H_2O = Uranyl(OH)_4^{2+} + 4H^{+}$	-32.4
$U^{+4} + H_2O = UOH^{+3} + H^{+}$	-0.541
$U^{+4} + 2H_2O = U(OH)_2^{+2} + 2H^{+}$	-1.091
$U^{+4} + 3H_2O = U(OH)_3^{+} + 3H^{+}$	-4.69
$U^{+4} + 4H_2O = U(OH)_4^{2+} + 4H^{+}$	-9.98
$U^{+4} + Sulfate^{2-} = USulfate^{+2}$	6.6
$Uranyl^{+2} + CO_3^{2-} = UranylCO_3$	9.94
$Uranyl^{+2} + 2CO_3^{2-} = Uranyl(CO_3)_2^{2-}$	16.61
$Uranyl^{+2} + 3CO_3^{2-} = Uranyl(CO_3)_3^{4-}$	21.84
$Ca^{+2} + Uranyl^{+2} + 3CO_3^{2-} = CaUranyl(CO_3)_3^{2-}$	27.18
$2Ca^{+2} + Uranyl^{+2} + 3CO_3^{2-} = Ca_2Uranyl(CO_3)_3$	30.7

Kinetic reactions are defined for sulfate bioreduction, Fe(III) bioreduction, U(VI) bioreduction, Fe(III) reduction by HS, U(VI) oxidation by Fe(III), and precipitation/dissolution of sulfur as follows:

Sulfate reduction (biotic)



Fe(III) reduction (biotic)



U(VI) reduction (combined abiotic and biotic, see text)



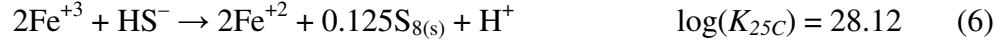
U(IV) oxidation by Fe(III)(abiotic):



HS⁻ oxidation by Fe(III) (abiotic):



These reactions are coupled to the precipitation/dissolution of solid phases: $\text{UO}_{2(\text{bio})}$, hematite, siderite, FeS , which are considered to be in equilibrium. However, the precipitation/dissolution of sulfur is expressed kinetically by the following reaction:



The sulfate and initial iron reduction reactions are modeled using a conventional dual-Monod rate law with biomass growth:

$$R_A = q C_b \frac{C_D}{k_D + C_D} \frac{C_A}{k_A + C_A} f_G \quad (7)$$

with

$$R_b = Y_b R_A - b C_b \quad (8)$$

$$f_G = (1 - Q/K) \quad (9)$$

where the subscripts A , D , and b represent electron acceptor, electron donor, and biomass, respectively, C is concentration, k is half saturation constant in units of C , q is the rate of maximum substrate utilization (in units of moles per time, per biomass), Y the microbial yield coefficient (in units of biomass per substrate), and b the cell decay rate (in units of per time).

The rates of reactions 4-6 are expressed as

$$R = r f_G, \quad (10)$$

where r (in units of moles per time) is assumed to be constant. Except for reaction 6, f_G is calculated from as $f_G = -(1 - K/Q)$, to ensure that f_G stays between -1 and 1.

For Parts II and III:

a. Flow model

Domain and Duration

- Dimension: 18 m in length, 10.5 m in width (see Figure 1)
- Grid discretization: 0.5m x 0.5m (see Figure 1)
- Duration: 60 days

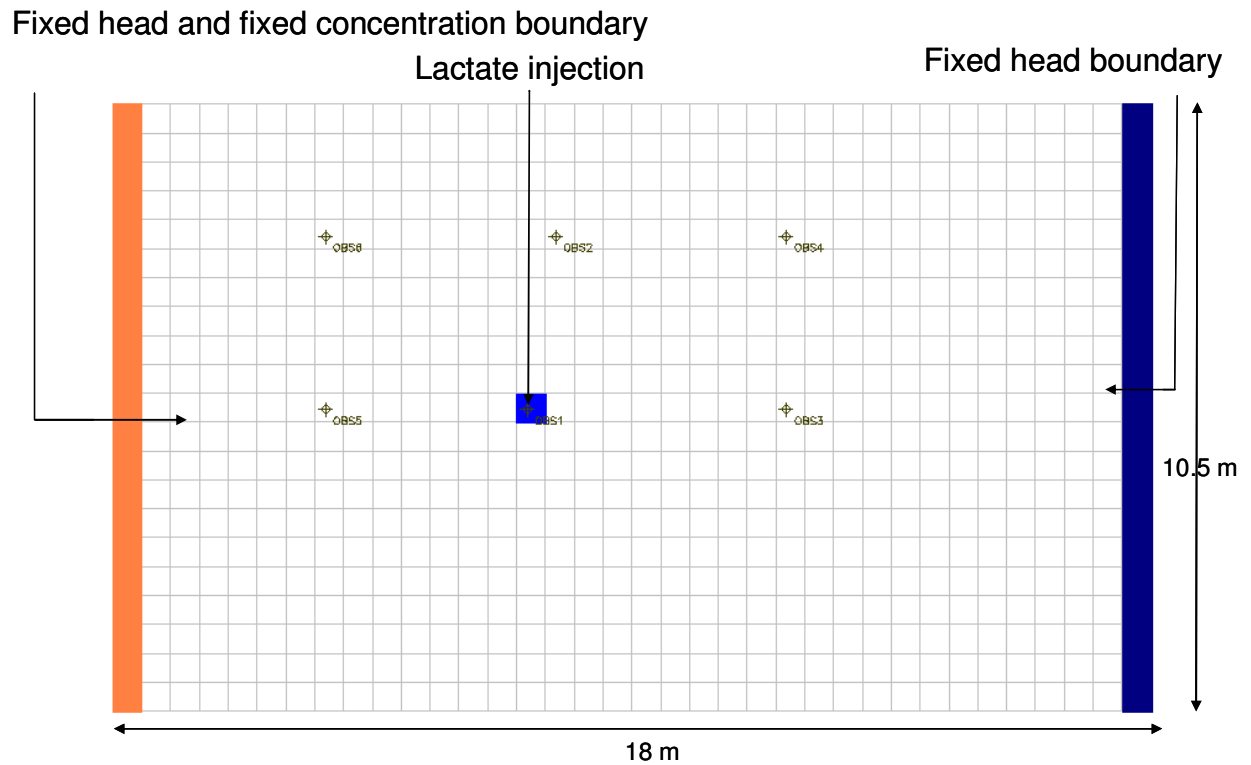


Figure 1. Model grid and boundary conditions

Material properties

- Hydraulic conductivity: spatially constant at 0.432 m/d

Initial conditions

- Steady state

Boundary conditions

- Specified head on west (8.75m) and east boundaries (8m).

b. Transport model

Material properties

- Longitudinal dispersivity : 1.0 m, Transverse dispersivity 0.1 m
- Porosity = 0.25

Initial conditions

- Initial aqueous concentrations:

Chemical component	Concentration (mol/L _{water})
Lactate	0
Acetate	0

Sulfate	0.02
S(-2)	0
Feii	0
Fe(3)	Equilibrium with hematite
U(6)	0.00009
Ca	0.0004
Mg	0.004
C(4)	0.0001558
U(4)	0
Na	0.0522
Cl	8.20E-04
pH	7.2

- Initial mineral concentrations:

Mineral	Concentration (mol/L_{water})
Hematite	7.25E-03
Uraninite	0
Sulfur	0
Mackinawite	0
Siderite	0

-Initial biomass: 3 g/L

Boundary conditions:

Constant concentration boundary at the left end of the domain with constant inflow of same U(VI) containing water at the domain. Temporary slow injection of lactate in the middle of the domain (see Figure 1) occurs with the following water compositions:

Chemical component	Concentration (mol/L_{water})	Concentration of injected water (mol/L_{water})
Lactate	0	0.03
Acetate	0	0
Sulfate	0.02	0.02
S(-2)	0	0
Feii	0	0
Fe(3)	Equilibrium with hematite	Equilibrium with hematite
U(6)	0.00009	0
Ca	0.0004	0.0004
Mg	0.004	0.004
C(4)	0.0001558	0.0001558
U(4)	0	0
Na	0.0522	0.08235
Cl	8.20E-04	8.20E-04
pH	7.2	7.2

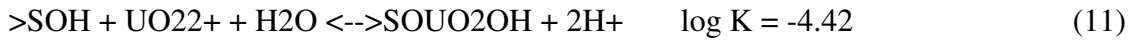
-Initial biomass: 3 g/L

c. Reaction Network Description:

-Part II: same as Part I

-Part III: same as Part I + surface complexation reactions of U(VI):

Non-electrostatic generalized surface complexation model for U(VI) adsorption for Hanford 300A sediment (Bond et al., 2008):



4. Model output comparison:

The following output should be compared in the benchmarking:

Part I:

Evolution with time of total aqueous concentrations of:

Lactate, Acetate, Sulfate, Fe(2), S(-2), Biomass, U(6)

Part II and Part III:

Evolution with time of total aqueous concentrations of:

Lactate, U(6) at the location of the injection well and 4.5 m down gradient of the injection well.

5. Model Input:

Calibrated kinetic parameters used for the simulation of U(VI) bioreduction in the presence of (separately) hematite:

Process	Reaction	Rate Law	q (mol/s/mg _{cells}) or r (mol/s)	k_D (mol/L)	k_A (mol/L)	Y_b (mg _{cells} /mol)	b (1/s)
Sulfate bioreduction	(1)	(7)	10^{-8}	2×10^{-2}	2×10^{-2}	1600	10^{-8}
Fe(III) bioreduction	(2)	(7)	10^{-11}	2×10^{-2}	10^{-20} (a)	1600	10^{-8}
U(VI) bioreduction	(3)	(17)	8×10^{-11}	4×10^{-2}			
Fe(III) reduction by HS	(5)	(16)	1.8×10^{-11}				
U(IV) oxidation by Fe(III)	(4)	(16)	2×10^{-11}				
Sulfur precipitation/dissol. ^(b)	(6)	(16)	1.4×10^{-11}				

^(a) Rate assumed essentially unlimited by the electron-acceptor (donor-limited experiments)

^(b) Rate incorporates (and assumes) a constant surface area

References

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- Sani R.K., Peyton B.M., Amonette J.E. and Geesey G.G. (2004) Reduction of uranium(VI) under sulfate-reducing conditions in the presence of Fe(III)-(hydr)oxides, *Geochim. Cosmochim. Acta* 68, 2639–2648.